Kinam Kim

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Research Interest

I am deeply interested in AI research, particularly in **Vision-Language-Action (VLA) models** [C1, C4] and **World Models** built on video generative models [C3, C5-C6, P1]. More recently I have been working on **coding agents for robotics** [O1], using them to produce the dexterous manipulation data that teleoperation and reinforcement learning struggle to provide. Across all three I am after the same thing: reliable robot behavior in the real world, built on experience that is generated rather than hand-collected.

Keywords: Vision-Language-Action (VLA) Models, World Models, Coding Agents for Robotics

Education

KAIST, Graduate School of AI — Daejeon, South Korea
M.S/Ph.D Integrated (Advisor: Prof. Jaegul Choo) — *Feb. 2025 - Present*
- GPA: 4.0 / 4.3

KAIST, School of Computing — Daejeon, South Korea
B.S. in Computer Science — *Mar. 2019 - Feb. 2024*
- GPA: 3.9 / 4.3

Samsung High School — Asan, South Korea
IT Diploma — *Mar. 2016 - Dec. 2018*

Work Experience

NVIDIA
Research Intern — *Jul. 2026 - Present*

Microsoft Research Asia — Tokyo, Japan
Research Intern — *Feb. 2026 - Jul. 2026*

KAIST DAVIAN Lab — Daejeon, South Korea
Undergraduate Research Intern — *Aug. 2023 - Mar. 2025*

ThetaOne Korea — Seoul, South Korea
Natural Language Processing AI Developer — *Jun. 2023 - Dec. 2023*

KAIST SiiT Lab — Daejeon, South Korea
Undergraduate Research Intern — *Jun. 2020 - Dec. 2020*

Teaching Assistant

KAIST AI Mentoring Program — Daejeon, South Korea
AI Mentor and Lead Program Coordinator — *Jun. 2020 - Dec. 2020*

Publications

Ongoing Project

[O1] **Scaling Dexterous Manipulation Data with Coding Agents**

Kinam Kim, Jaegul Choo

Ongoing research project, 2026

TL;DR: Teleoperation and reinforcement learning both scale poorly for multi-fingered hands, so we let a coding agent write the policy instead: it composes trajectory optimization, motion planning and scripted control, then revises its own code from simulation feedback. This yields bimanual dexterous demonstrations on contact-rich tasks such as Rubik's cube turning, carabiner clipping and bolt fastening, with no human demonstration at any point.

Conference Paper

[C1] **Object-Centric Residual RL for Zero-Shot Sim-to-Real VLA Enhancement**

Kinam Kim, Namiko Saito, Heecheol Kim, Katsushi Ikeuchi, Jaegul Choo, Yasuyuki Matsushita

Conference on Robot Learning (CoRL), 2026

Work done during an internship at Microsoft Research Asia

TL;DR: We propose an object-centric residual reinforcement learning framework that augments a base VLA policy with a residual policy trained in simulation. Our approach transfers zero-shot to the real world, substantially improving success rates on manipulation tasks such as pick-and-place and drawer closing.

[C2] **FlashSAC: Fast and Stable Off-Policy Reinforcement Learning for High-Dimensional Robot Control**

Donghu Kim*, Youngdo Lee*, Minho Park, **Kinam Kim**, I Made Aswin Nahendra, Takuma Seno, Sehee Min, Daniel Palenicek, Florian Vogt, Danica Kragic, Jan Peters, Jaegul Choo, Hojoon Lee

Robotics: Science and Systems (RSS), 2026

Outstanding Paper Award

TL;DR: FlashSAC sharply cuts the number of gradient updates in Soft Actor-Critic and compensates with larger models and higher data throughput, bounding weight, feature and gradient norms so the critic stays stable at that scale. It beats PPO and strong off-policy baselines on over 60 tasks across 10 simulators, and brings sim-to-real humanoid locomotion training down from hours to minutes.

[C3] **EgoX: Egocentric Video Generation from a Single Exocentric Video**

Taewoong Kang*, **Kinam Kim***, Dohyeon Kim*, Minho Park, Junha Hyung, Jaegul Choo

The IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2026

TL;DR: We explore generating egocentric video from single exocentric video for VR/AR applications. This capability allows third-person videos to be transformed into first-person perspectives, enabling users to step into the scene and experience it as if they were the subject.

[C4] **ACG: Action Coherence Guidance for Flow-based VLA models**

Minho Park*, **Kinam Kim***, Junha Hyung, Hyojin Jang, Hoijeong Jin, Jooyeol Yun, Hojoon Lee, Jaegul Choo

IEEE International Conference on Robotics and Automation (ICRA), 2026

TL;DR: We introduce Action Coherence Guidance (ACG), a training-free test-time guidance method that reduces noise-induced jerks, pauses, and jitter in VLA model policies. We show that ACG improves action coherence and significantly boosts success rates in RoboCasa, DexMimicGen, and real-world SO-101 manipulation tasks.

[C5] **Temporal In-Context Fine-Tuning for Versatile Control of Video Diffusion Models**

Kinam Kim*, Junha Hyung*, Jaegul Choo

The Thirty-Eighth Annual Conference on Neural Information Processing Systems (**NeurIPS**), 2025

TL;DR: We propose Temporal In-Context Fine-Tuning (TIC-FT), which inserts noisy buffer frames to adapt pretrained video diffusion models for controllable generation. We show that TIC-FT works with as few as 10–30 samples and improves both fidelity and efficiency over existing methods.

[C6] **Spatiotemporal Skip Guidance for Enhanced Video Diffusion Sampling**

Junha Hyung*, **Kinam Kim***, Susung Hong, Min-Jung Kim, Jaegul Choo

The IEEE/CVF Conference on Computer Vision and Pattern Recognition (**CVPR**), 2025

TL;DR: We propose Spatiotemporal Skip Guidance (STG), a training-free sampling method that enhances video diffusion models by simulating a weak model through selective layer skipping. We show that STG improves sample quality without sacrificing diversity or motion, unlike conventional CFG or autoguidance.

Preprints

[P1] **Cross-Frame Representation Alignment for Fine-Tuning Video Diffusion Models**

Sungwon Hwang*, Hyojin Jang*, **Kinam Kim**, Minho Park, Jaegul Choo

arXiv preprint, 2025

TL;DR: We propose Cross-frame Representation Alignment (CREPA), a regularization method that leverages neighboring frames to improve semantic consistency in video diffusion models. We show that CREPA boosts both visual fidelity and cross-frame coherence when fine-tuned with parameter-efficient methods like LoRA.

[P2] **Overcoming Camera-Facing Gaze Bias in EG3D Scene Generation**

Kinam Kim, Juhyun Lee, Youngdo Lee

arXiv preprint, 2023

TL;DR: We address the gaze-following problem in 3D GANs by integrating a pretrained gaze estimation model (L2CS-Net) into the EG3D training pipeline. We show that this fusion preserves realistic image quality while successfully correcting gaze direction on the FFHQ dataset.

Reference

Jaegul Choo — Seongnam, South Korea

Ph.D. Advisor (KAIST AI) — *Mar. 2021 - Present*

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- website

Yu-Chiang Frank Wang

Internship Manager (Research Director, Deep Learning & Computer Vision, NVIDIA) — *Jul. 2026 - Present*

- E-mail: ycwang@ntu.edu.tw
- website

Yasuyuki Matsushita — Tokyo, Japan

Internship Manager (Senior Director, Microsoft Research Asia - Tokyo) — *Feb. 2026 - Jul. 2026*

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